CHROMATIC HOMOTOPY THEORY SEMINAR MÜNSTER

Disclaimer: You can follow these notes during the talk or extend them or prepare your own. Essential references and content to be covered are always written at the beginning of each talk. All gray text is optional and can be easily skipped as it only provides some additional or background material which simplifies the understanding (in the author's opinion).

1. Talk 1. Why do formal group laws arise?

- Define Chern characteristic classes, discuss operations on the set of vector bundles. [1], [2]
- State the existence of the classifying space $BU(n)$ and show a sketch of the computation of its cohomology.
- Define complex-oriented and even cohomology theories, explain their importance.
- Work out two main examples: complex K-theory and integral cohomology, explain how formal group laws arise in these two examples.
- Define a formal group law.

While the development of the theory of characteristic classes was motivated by the study of linearly independent vector fields on manifolds, since then this theory (and a lot of mathematics) has found its description in the purely homotopical language, which I provide below.

Definition 1.1. A *complex vector bundle* ω of dimension n over X consists of a topological space E and a projection map $\pi\colon E \to X$, together with the structure of a complex vector space in each fibre $\pi^{-1}(x)$, subject to the following local triviality condition:

Each point of X possesses a neighborhood U such that $\pi^{-1}(U)$ is homeomorphic to $U \times \mathbb{C}^n$ under a homeomorphism which maps each fiber $\pi^{-1}(x)$ complex linearly to $\{x\} \times \mathbb{C}^n$.

One reason for the importance of complex vector bundles is that they are represented by a certain classifying space, precisely:

Theorem 1.2. *For each $n \in \mathbb{N}$ there exists a topological space $BU(n)$ such that for any paracompact Hausdorff space X there is a natural bijection*

$$[X, BU(n)] \cong \left\{ \begin{matrix} \textit{isomorphism classes of complex vector} \\ \textit{bundles of rank } n \textit{ over } X \end{matrix} \right\}.$$

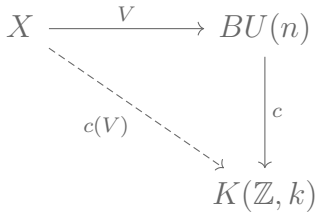
Remark 1.3. The space $BU(n)$ is a classifying space for the topological group $U(n)$ – the group of linear isometries of $\mathbb{C}^n$.

Another representability result is that the cohomology functor is represented by Eilenberg–MacLane spaces, which, as all of you know, is called the Brown representability theorem.

These two ingredients allow us to have a quite concrete description of certain cohomology classes in, in general, quite complicated cohomology groups $H^*(X, \mathbb{Z})$.

Thus, for a fixed $c \in [BU(n), K(\mathbb{Z}, k)]$ and any vector bundle $V \to X$ of rank n we can define a cohomology class $c(V) \in H^k(X, \mathbb{Z})$ by just composing:

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Here we slightly abuse notation and write V for the corresponding classifying map $X \to BU(n)$ of the vector bundle V.

As it is defined, this assignment is too arbitrary; it lacks some naturality properties.

Definition 1.4. An integral complex characteristic class p is an assignment of a singular cohomology class

$$p(V) \in \mathrm{H}^n(X, \mathbb{Z})$$

to each complex vector bundle $V \to X$ which is natural with respect to pullback of vector bundles. That is, for any continuous map $f\colon Y \to X$ of topological spaces we have an equality

$$f^* p(V) = p(f^* V)$$

of classes in $\mathrm{H}^n(Y, \mathbb{Z})$.

The fundamental example of complex characteristic classes is given by the *Chern classes*

$$c_k(V) \in \mathrm{H}^{2k}(X, \mathbb{Z}),$$

which are uniquely determined by a few simple axioms. To state these, it is convenient to introduce the *total Chern class* given by

$$c(V) := 1 + c_1(V) + c_2(V) + \dots.$$

Using this notation, the axioms satisfied by the Chern classes are as follows:

(1) $c(f^*V) = f^* c(V)$ (Naturality)
(2) $c(V \oplus W) = c(V) \cup c(W)$ (Whitney sum)
(3) $c(\gamma_1) = 1 + t$, where $\gamma_1 \to \mathbb{CP}^\infty \cong BU(1)$ is the tautological line bundle and $t \in \mathrm{H}^2(BU(1), \mathbb{Z})$ is a generator (Normalization)

These axioms give a theory that is as elegant as possible, in the following sense:

(1) the Chern classes are uniquely defined by the above axioms and
(2) they generate all integral complex characteristic classes; that is, any other class is a polynomial in Chern classes. It will become clear what this means later, see Proposition 1.5 below.

One might ask why we define the k-th Chern class to be in the $2k$-th cohomology group. The answer is the following computation of the cohomology of the classifying space $BU(n)$:

Proposition 1.5. *The cohomology ring of the classifying space $BU(n)$ is given by*

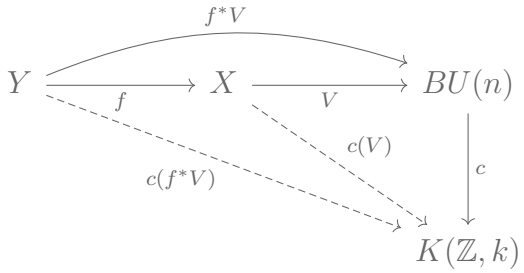
$$H^*(BU(n), \mathbb{Z}) \cong \mathbb{Z}[c_1, c_2, \dots, c_n],$$

where c_k is a generator in degree $2k$.

Thus, all odd cohomology groups of $BU(n)$ just vanish, and therefore there are no non-trivial characteristic classes in odd degrees.

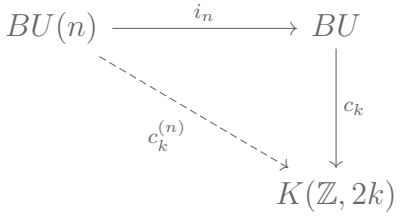
Let us also discuss how to construct characteristic classes satisfying the aforementioned axioms.

First, note that from our construction of Chern classes, given by composing two classifying maps, the first property (naturality with respect to pullbacks) is automatically satisfied:



To have some compatibility with respect to the cup product, we want Chern classes of different dimensions to be compatible with respect to the inclusion $U(n) \hookrightarrow U(n+1)$. This is achieved by considering the infinite unitary group $U = \bigcup_{n\geq 0} U(n)$ and its classifying space $BU = \bigcup_{n\geq 0} BU(n)$.

Let us choose characteristic classes $c_k \in [BU, K(\mathbb{Z}, 2k)]$ and then restrict other Chern classes along natural inclusions $BU(n) \hookrightarrow BU$:



Usually the upper index n is omitted, and we just write c_k for the Chern class of a rank n vector bundle.

Finally, the normalization axiom, since the cohomology ring $H^*(BU(1), \mathbb{Z}) \cong \mathbb{Z}[t]$ is generated by a single element t in degree 2, we are choosing $c_0^{(1)}$ to be one and $c_1^{(1)}$ to be the generator t.

The question of classifying characteristic classes can be reduced to determining cohomology of certain topological spaces. Namely, for each $n \geq 1$ there exists a certain topological space

$$BU(n) := \{V \subset \mathbb{C}^\infty \mid \dim(V) = n\},$$

the *classifying space of the unitary group* $U(n)$.

Explicitly, as written above, the model for $BU(n)$ is the space of n-dimensional linear subspaces of $\mathbb{C}^\infty$. The importance of this topological space comes from the fact that it carries a particularly important vector bundle.

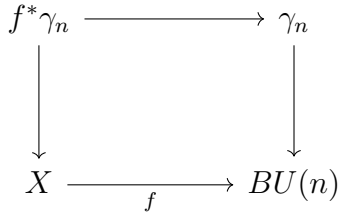
Definition 1.6. The tautological vector bundle $\gamma_n \to BU(n)$ is the vector bundle

$$\gamma_n := \{(V, x) \in BU(n) \times \mathbb{C}^\infty \mid x \in V\},$$

whose fiber over a point corresponding to $V \subset \mathbb{C}^\infty$ is the vector space V itself.

The tautological vector bundle has the following universal property: for paracompact Hausdorff space X, there is a canonical bijection given by pulling back the tautological bundle γ_n:

$$[X, BU(n)] \longrightarrow \{\text{rank } n \text{ vector bundles over } X\}_{/\text{isomorphism}}.$$



In fact, the tautological bundle γ_n, being a rank n vector bundle over $BU(n)$, corresponds to a classifying map $BU(n) \to BU(n)$, namely the identity.

Therefore, the correspondence above might seem tautological (sorry for the pun): since pulling back along f corresponds to precomposition with f, it says that precomposing $\mathrm{id}_{BU(n)}$ with f gives back f, and hence the same vector bundle.

The theorem is certainly not obvious: above, I used it circularly before proving it.

Proposition 1.7. *The cohomology of the classifying space*

$$H^*(BU(n), \mathbb{Z}) \simeq \mathbb{Z}[[c_1(\gamma_n), \ldots, c_n(\gamma_n)]]$$

is isomorphic to the power series ring in the Chern classes, where $|c_k(\gamma_n)| = 2k$.

In particular, since $BU(1) \cong \mathbb{C}P^\infty$ we have:

$$H^*(BU(1), \mathbb{Z}) \cong H^*(\mathbb{C}P^\infty, \mathbb{Z}) \cong \mathbb{Z}[[c_1(\gamma_1)]] \cong \mathbb{Z}[[t]]$$

In fact, the whole calculation above boils down to what happens for $\mathbb{CP}^\infty$, and we'll make it explicit now. More precisely, suppose we have chosen a generator

$$t \in H^2(\mathbb{CP}^\infty, \mathbb{Z}),$$

corresponding to a choice of the first Chern class of the tautological line bundle. Any such choice determines by a standard calculation an isomorphism

$$H^*(\mathbb{CP}^\infty, \mathbb{Z}) \simeq \mathbb{Z}[[t]]$$

with the power series ring in variable $t = c_1(\gamma_1)$.

Warning 1.1. The appearance of the power series ring can be confusing at first, so note that the rings in the above isomorphism are graded and we claim that they're isomorphic in this sense; that is, the subspaces of homogeneous elements are isomorphic.

Since $\mathbb{Z}$ is concentrated in a single degree, the two expressions $\mathbb{Z}[[t]]$ and $\mathbb{Z}[t]$ denote the same graded ring (because if we fix a degree, the homogeneous elements are the same), and it is customary in many textbooks to use the latter notation. This works for singular cohomology $H\mathbb{Z}$; for more general cohomology theories, though, this really will be a power series ring and not the polynomial one.

In the axioms of Chern classes we used only one natural operation on vector bundles, namely the direct sum. In fact, there is substantially more structure on (the groupoid of) vector bundles.

There is a natural monoidal structure: given two vector bundles $V, W \to X$ we can form their tensor product $V \otimes W$. Geometrically this is done by forming tensor products of the fibres over each point $x \in X$ and its respective open neighborhoods; see [3].

In fancier words, the space $\coprod_{n\in\mathbb{N}} BU(n)$ possesses the structure of an E_∞-semiring space, where (on the level of homotopy category):

- the addition is given by the map classifying direct sum $BU(n) \times BU(m) \to BU(n+m)$
- the multiplication is given by the map classifying tensor product $BU(n) \times BU(m) \to BU(nm)$

Units are coming from the maps $pt = BU(0) \to BU(0)$ for addition and (any of the maps) $pt = BU(0) \to BU(1)$ for multiplication, for example, the one classifying the trivial line bundle.

Furthermore, the concrete geometric model of these classifying maps comes from the block sum of matrices $U(n) \times U(m) \to U(n+m)$ and the tensor product of matrices $U(n) \times U(m) \to U(nm)$, respectively.

Remark 1.8. Crossover: *Using this tensor product we show that E_∞-(semi)ring objects in An give rise to E_∞-ring spectrum objects i.e. in Sp^{cn}* [4]. As an application one obtains the E_∞ spectrum

ku, which is the connective cover of the spectrum KU that represents topological complex K-theory mentioned in the first talk of the K-theory seminar [5].

Sketch of proof of Proposition 1.7. There's a well-defined homotopy class of maps

$$BU(1)^{\times n} \cong (\mathbb{CP}^\infty)^n \longrightarrow BU(n)$$

classifying the sum of the pullbacks of the tautological bundles over the factors, which is a vector bundle of rank n. This sum does not depend on the order of the factors, and so this map is equivariant with respect to the Σ_n-action on the source permuting the factors. Thus, we get a map

$$H^*(BU(n), \mathbb{Z}) \longrightarrow H^*((\mathbb{CP}^\infty)^n, \mathbb{Z})^{\Sigma_n} \cong \mathbb{Z}[[t_1, \dots, t_n]]^{\Sigma_n}$$

into Σ_n-fixed points, where $t_i := c_1(\pi_i^* \gamma_1)$ is the first Chern class of the pullback of the tautological line bundle from the i-th coordinate. The action of Σ_n permutes the t_i's and one can show that the above ring map is an isomorphism.

This implies that the cohomology of $BU(n)$ is isomorphic to the ring of symmetric power series in t_i's; it is well-known that such a ring is itself a power series ring in the elementary symmetric polynomials.

For example, we have

$$e_1 := t_1 + \cdots + t_n \in \mathbb{Z}[[t_1, \dots, t_n]]^{\Sigma_n},$$

while the higher ones are described by slightly more involved formulas. These are the power series generators of the cohomology of $BU(n)$ visible in Proposition 1.7. That is, $c_k(\gamma_n)$ corresponds to the k-th elementary symmetric polynomial. □

Question. *What happens if instead of $H^*(-, \mathbb{Z})$ we consider characteristic classes valued in a different multiplicative cohomology theory $E^*(-)$?*

As it turns out, as long as $\mathbb{CP}^\infty$ behaves E-cohomologically the same way it does with respect to integral cohomology, the whole story works out in the same way. To be more precise let us formally give a definition which we alluded to before.

Definition 1.9. A *complex orientation* for a multiplicative cohomology theory E is a choice of an element

$$t \in \widetilde{E}^2(\mathbb{CP}^\infty)$$

which under the map

$$\widetilde{E}^2(\mathbb{CP}^\infty) \longrightarrow \widetilde{E}^2(\mathbb{CP}^1) \cong \widetilde{E}^2(S^2) \cong E^0(pt)$$

restricts to the unit $1 \in E^0(pt)$. A multiplicative cohomology theory is *complex orientable* if it admits a complex orientation.

Definition 1.10. The 0-th complex K-theory of an anima X is $KU^0(X) := [X, \mathbb{Z} \times BU]$. For compact Hausdorff X this agrees with the Grothendieck group of $(\mathrm{Vect}_{\mathbb{C}}(X), \oplus)$.

The negative part is defined, for $n \geq 0$, as $KU^{-n}(X) := [X, \Omega^n(\mathbb{Z} \times BU)]$.

By Bott periodicity $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$, so $KU^{-n-2} \cong KU^{-n}$, and we set $KU^n := KU^{n-2}$ for $n > 0$.

Definition 1.11. A cohomology theory E is called *even* if $E^* \coloneqq E^*(*)$ is concentrated in even degrees.

Example 1.12. For example, E can be given by complex K-theory KU, since

$$KU^* \cong \mathbb{Z}[\beta^{\pm 1}],$$

where β is the Bott element of degree -2.

$$KU^{-2}(*) \cong [*, \Omega^2(\mathbb{Z} \times BU)] \cong \pi_2(BU) \cong [\text{Hurewicz}] \cong H^2(BU, \mathbb{Z}) \cong \mathbb{Z}.$$

The even cohomology theories are particularly important since they provide a natural source of complex oriented cohomology theories. Indeed, assume that E is an even cohomology theory. Then, all of the maps

$$E^*(\mathbb{CP}^{n+1}) \longrightarrow E^*(\mathbb{CP}^n)$$

are in fact surjective, so we can lift any class in $\mathbb{CP}^1$ to $\mathbb{CP}^\infty$.

The surjectivity follows from the existence of the cofiber sequence of the form $\mathbb{CP}^n \to \mathbb{CP}^{n+1} \to S^{2n+2}$ by two steps:

(1) Apply the long exact sequence in cohomology for the cofiber sequence above
(2) Note that $\tilde{E}^q(S^{2n+2}) = E^{q-2n-1}(pt)$ vanishes for even q, so there are no obstructions to lifting from $E^q(\mathbb{CP}^n)$ to $E^q(\mathbb{CP}^{n+1})$, while for odd q we have $E^q(\mathbb{CP}^n) = 0$ by the standard induction argument.

Recall Warning 1.1 above; spoiler: $KU^*(\mathbb{CP}^\infty) \cong KU^*(*)[[t]]$, with $|t| = 2$ and $|\beta| = -2$.

Therefore, here we can't replace the power series ring by a polynomial one, since already in degree zero we have elements of the form $\sum_{i=0}^\infty \beta^i t^i$ which are not polynomials in t.

For complex-oriented cohomology theories, the story of Chern classes works out the same way as integrally.

Proposition 1.13. *If E is complex-orientable, then*

$$E^*(\mathbb{CP}^\infty) \cong E^*[[t]]$$

for any choice of complex orientation t.

One can then carry out the remaining calculations of the cohomology of classifying spaces. That is, if we set $c_1(\gamma_1) = t$, then we will have

$$E^*(BU(n)) \cong E^*[[c_1(\gamma_n), \dots, c_n(\gamma_n)]]$$

for unique Chern classes satisfying the three axioms given above, namely naturality, Whitney sum and normalization.

So far we have taken into account only the additive structure on the set of vector bundles. Thus, there is a natural question of what happens if we also take into account the multiplicative structure given by the tensor product.

Question. *Given a complex-oriented cohomology theory E, what is the formula $F_E(t_1, t_2)$ such that for any pair of line bundles L, K over a topological space X we have*

$$c_1(L \otimes K) = F_E(c_1(L), c_1(K))$$

as elements of $E^2(X)$?

The existence of the formula can be checked by reducing to the universal case, using the Yoneda lemma. Namely, the tensor product of line bundles is classified by a homotopy class of maps

$$m \colon BU(1) \times BU(1) \longrightarrow BU(1)$$

which in turn induces a map on cohomology:

$$m^* \colon E^*(BU(1)) \longrightarrow E^*(BU(1) \times BU(1))$$

which under the identification $E^*(BU(1)) \cong E^*[[t]]$ and the Künneth isomorphism is of the form

$m^*\colon E^*[[t]] \longrightarrow E^*(BU(1)\times BU(1)) \cong$ [this iso comes from a computation which uses complex orientability]

where t_i are the pullbacks of t along the projections onto the two factors of $BU(1) \times BU(1)$.

Taking pullbacks we see that the formula for the first Chern class of a tensor product is given by $F_E(t_1, t_2) = m^*(t)$. As it turns out, this formula *does* fundamentally depend on the choice of the cohomology theory and the choice of an orientation.

Remark 1.14. A very concrete description of the map $\mathbb{C}P^\infty \times \mathbb{C}P^\infty \to \mathbb{C}P^\infty$ looks as follows:

$$[\ldots : a_i : \ldots] \times [\ldots : b_j : \ldots] \to [\ldots : \sum_{i+j=k} a_i b_j : \ldots]$$

Example 1.15. When $E = H\mathbb{Z}$ is integral cohomology, then

$$F_{H\mathbb{Z}}(t_1, t_2) = t_1 + t_2.$$

Example 1.16. Let $E = KU$ be complex K-theory. With the complex orientation $t = \beta^{-1}([\mathcal{O}(1)] - 1)$, we have

$$F_{KU}(t_1, t_2) = \beta^{-1}\big((\beta t_1 + 1)(\beta t_2 + 1) - 1\big).$$

By the universality argument above, the promised formula is always a power series in two variables (in the two cases above it happens to be a polynomial, but that is not usually the case); however, it is not completely arbitrary. Since the tensor product of line bundles is commutative, unital and associative, we must in turn have that

(1) $F_E(t_1, t_2) = F_E(t_2, t_1)$,
(2) $F_E(0, t) = F_E(t, 0) = t$,
(3) $F_E(F_E(t_1, t_2), t_3) = F_E(t_1, F_E(t_2, t_3))$.

A power series over a ring (in this case E^*) satisfying these three identities is called a *formal group law*. These are certain objects of algebro-geometric nature whose study will be one of the main goals of this course.

The above construction, which is due to Quillen, produces an association

$$\{\text{complex-oriented cohomology theories}\} \longrightarrow \{\text{formal group laws}\}.$$

This correspondence is not quite one-to-one, but it is highly non-trivial:

(1) many important classes of formal group laws can be shown to come from a complex-oriented cohomology theory and
(2) the associated formal group law determines many of its properties.

Importantly, formal group laws do not just form a discrete set, but come with a natural notion of isomorphism. Thus, they can be assembled into what algebraic geometers call the *moduli stack of formal groups*. As an informal slogan, the geometry of this moduli stack controls the behaviour of stable homotopy theory, and the above correspondence is just one of the instances of this phenomenon.

References

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